Estimation of Multiple Treatment Effects on Software Revenue

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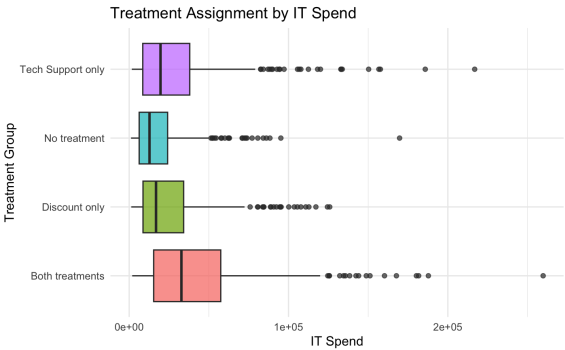
Problem Setup

Data about a software company that offers two types of interventions to other companies in order to increase revenue.

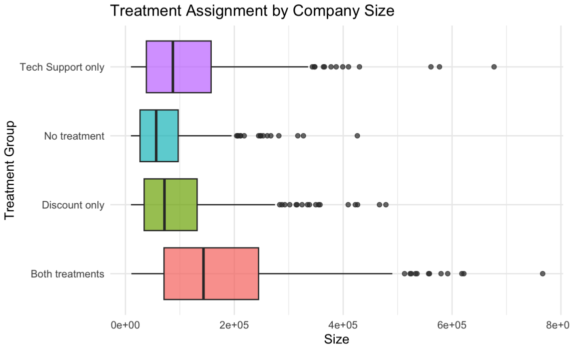
- Outcome: Revenue
- Treatments:
  - Discount and Tech Support
  - 4 treatment arms: Discount only, Tech support only, Both and No treatment.
- 8 Covariates about customer characterstics, like size and IT spend
- Goal: estimate causal effect of these interventions on revenue

IT spend is not balanced



- High IT spend firms receive more treatments

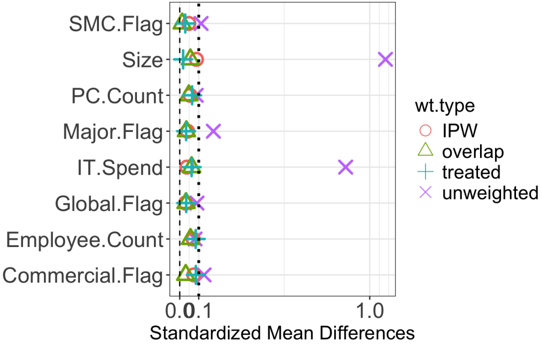
Size is not balanced



- Larger firms receive more treatments

Treatment groups are not comparable

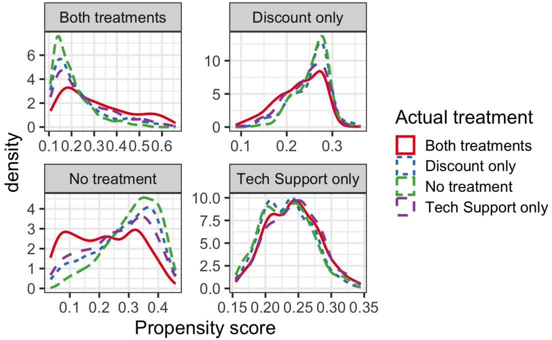
- Strong covariate imbalance across groups



- Unweighted comparisons are biased
- Propensity score weighting can help to make the groups comparable

Propensity score weighting

- Methods:
  - IPW, Overlap, ATT, Doubly Robust
- Trimming removes extreme propensity scores
- Overlap looks decent



- DR estimator $\rightarrow$ lowest variance

Result: Both treatments vs No treatment

- Want to estimate
  $\tau = \mathbb{E}\big[Y(\text{Both treatments}) - Y(\text{No treatment})\big]$
- Unweighted (naive) comparison: \$20,198
- Trimmed DR estimation: \$11,810

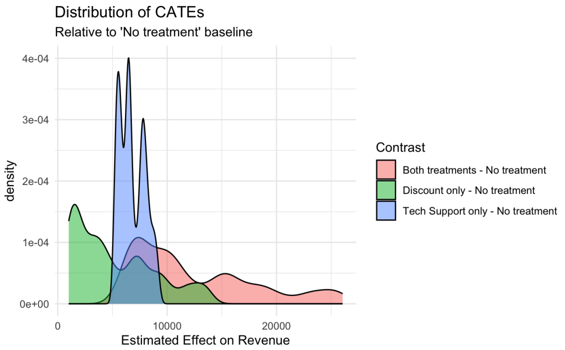
- Ignoring confounding variables $X \rightarrow$ overestimation

Method	Trimmed	Estimate	Std. Error
IPW	Untrimmed	13405.60	403.30
IPW	Trimmed	12222.98	219.14
Overlap	Untrimmed	12066.75	255.03
ATT	Untrimmed	12785.37	465.34
DR	Untrimmed	12917.68	189.91
DR	Trimmed	11810.44	134.87

Table: Contrast 2: $\mathbb{E}\big[Y(\text{Both treatments}) - Y(\text{No treatment})\big]$

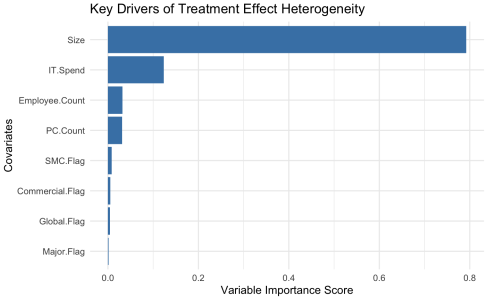
Causal forest to estimate CATE

- Causal forest model shows strong heterogeneity across customers:



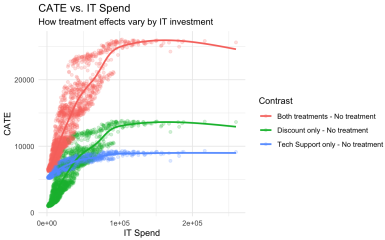
What Drives Heterogeneity?

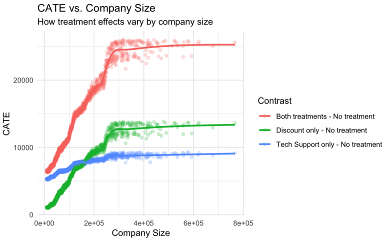
- Causal forest feature importance:



- Most important:
  - Company size
  - IT spend
- These were also the most unbalanced covariates

CATE Patterns





- Treatment effects are larger for firms with higher IT spend and larger size.

Treatment allocation optimization

- The causal forest model predictions allows us to further optimize treatment allocation based on these heterogeneous effects.
- Treatment Cost Assumptions:
  - Discount: \$6000
  - Tech support: \$7000
  - Both: \$13000
  - No treatment: \$0

Let $\tau_k(X) = \mathbb{E}[Y(k) - Y(0) \mid X]$, and c_k the cost of treatment k,

$$\text{net}_k(X) = \tau_k(X) - c_k$$

Let $d_{ik} = 1$ if unit i is assigned to treatment k, and 0 otherwise. B is the total budget:

$$\arg\max_{d_{ik}} \sum_{i=1}^{n} \sum_{k} d_{ik}\, \text{net}_k(X_i), \quad \text{s.t.} \quad \sum_{k} d_{ik} = 1, \ \sum_{i,k} d_{ik} c_k \leq B$$

Allocation Results

- Estimated incremental revenue (current assignment):

  $1,225,866

- Estimated incremental revenue (optimized assignment):

  $4,307,291

- **+ $3.08M improvement** under the same budget
- Gains driven by targeting high-impact customers using causal inference